Category	Part	Model & Link	Price	Quantity	Total
MCU	High Level Controller	Raspberry Pi 5 8GB	80	1	80
MCU	Low Level Controller	Teensy 4.0	23.8	1	23.8
MCU	Pi Active Cooler	SC1148	5	1	5
MCU	SD Card	SanDisk 64GB	10.9	1	10.9
UI	Pi Touchscreen	SC1227	60	1	60
Sensing	LiDAR	RPLIDAR A2M12	209	1	209
Sensing	Camera	RPi Camera	37.9	1	37.9
Sensing	UltraSonic Ranger	101020010	4.3	4	17.2
Sensing	IMU	Adafruit 2472	34.95	1	34.95
Sensing	Base Hat	103030275	10.9	1	10.9
Drivetrain	Motor Driver	4 Pack L298N	20	1	20
Drivetrain	Motor with Encoder	FIT0186	16.5	2	33
Power	Battery	Talentcell 26400mAh	63.99	1	63.99
Power	Logic Level Shifter	Adafruit 5637	2.95	3	8.85
Power	Buck Converter	UBEC Step Down	9.95	1	9.95
Power	Fuses	16 AWG Inline Fuse	8.99	1	8.99
Power	Barrel to Screw Terminal	DC Barrel Connector	5.99	1	5.99
Power Hub	Distribution Board	XT60 PDB + 90A traces	9.6	1	9.6
Net Total	**650.02**		**Total + Tax (8%)**		**702.0216**

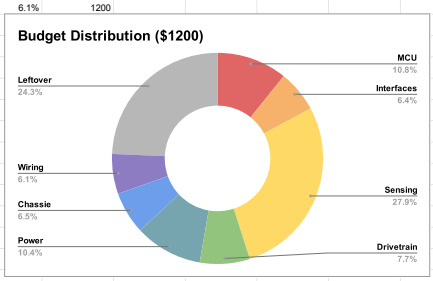
Category	Part	Model & Link	Price	Quantity	Total
MCU	High Level Controller	Raspberry Pi 5 8GB	80	1	80
MCU	Low Level Controller	Teensy 4.0	23.8	1	23.8
MCU	Pi Active Cooler	SC1148	5	1	5
MCU	SD Card	SanDisk 64GB	10.9	1	10.9
UI	Pi Touchscreen	SC1227	60	1	60
Sensing	LiDAR	RPLIDAR A2M12	209	1	209
Sensing	Camera	RPi Camera	37.9	1	37.9
Sensing	UltraSonic Ranger	101020010	4.3	4	17.2
Sensing	IMU	Adafruit 2472	34.95	1	34.95
Sensing	Base Hat	103030275	10.9	1	10.9
Drivetrain	Motor Driver	4 Pack L298N	20	1	20
Drivetrain	Motor with Encoder	FIT0186	16.5	2	33
Power	Battery	Talentcell 26400mAh	63.99	1	63.99
Power	Logic Level Shifter	Adafruit 5637	2.95	3	8.85
Power	Buck Converter	UBEC Step Down	9.95	1	9.95
Power	Fuses	16 AWG Inline Fuse	8.99	1	8.99
Power	Barrel to Screw Terminal	DC Barrel Connector	5.99	1	5.99
Power Hub	Distribution Board	XT60 PDB + 90A traces	9.6	1	9.6
Net Total	**650.02**		**Total + Tax (8%)**		**702.0216**

Main Items					
Category	**Part**	**Model & Link**	**Price**	**Quantity**	**Total**
MCU	High Level Controller	Raspberry Pi 5 8GB	$80.00	1	$80.00
MCU	Low Level Controller	Teensy 4.0	$23.80	1	$23.80
UI	Pi Touchscreen	SC1227	$60.00	1	$60.00
Sensing	LiDAR	RPLIDAR A2M12	$209.00	1	$209.00
Sensing	Camera	RPi Camera	$37.90	1	$37.90
Sensing	UltraSonic Ranger	101020010	$4.30	4	$17.20
Sensing	IMU	Adafruit 2472	$34.95	1	$34.95
Drivetrain	Motor Driver	4 Pack L298N	$20.00	1	$20.00
Drivetrain	Motor with Encoder	FIT0186	$16.50	2	$33.00
Power	Battery	Talentcell 26400mAh	$63.99	1	$63.99
Other Items					$261.05
Net Total	$840.89		Total + Tax (8%)		$908.16

$261.05	$579.84
	840.89

Category	Part	Model & Link	Price	Quantity	Total
MCU	High Level Controller	Raspberry Pi 5 8GB	80	1	80
MCU	Low Level Controller	Teensy 4.0	23.8	1	23.8
MCU	Pi Active Cooler	SC1148	5	1	5
MCU	SD Card	SanDisk 64GB	10.9	1	10.9
Interfaces	Pi Touchscreen	SC1227	60	1	60
Sensing	LiDAR	RPLIDAR A2M12	209	1	209
Sensing	Camera	RPi Camera	37.9	1	37.9
Sensing	UltraSonic Ranger	101020010	4.3	4	17.2
Sensing	IMU	Adafruit 2472	34.95	1	34.95
Sensing	Base Hat	103030275	10.9	1	10.9
Drivetrain	Motor Driver	4 Pack L298N	20	1	20
Drivetrain	Motor with Encoder	FIT0186	16.5	2	33
Power	Battery	Talentcell 26400mAh	63.99	1	63.99
Power	Logic Level Shifter	Adafruit 5637	2.95	3	8.85
Power	Buck Converter	UBEC Step Down	9.95	1	9.95
Power	Fuses	16 AWG Inline Fuse	8.99	1	8.99
Power	Barrel to Screw Terminal	DC Barrel Connector	5.99	1	5.99
Power	Distribution Board	XT60 PDB + 90A traces	9.6	1	9.6
Chassie	Aluminum Extrusions	4pc 400mm	18.99	1	18.99
Chassie	Extrusion Nuts and Bolts	200pc	9.99	1	9.99
Drivetrain	Wheels & Hubs (90mm rubber)	Rubber Wheel	2	12	24
Drivetrain	Caster Wheel (50mm swivel)	Third-point balance suppo	1	8	8
Power	Master Power Switch (High-Current)	Main system on/off switc	1	8	8
Chassie	3D fillament	PETG 2KG	1	24.99	24.99
Chassie	12.6V Li-ion Smart Charger	Battery recharging unit	1	18	18
Wiring	Zip Ties & Cable Clips	Cable organization and st	1	6	6
Wiring	Signal Wire (22–24 AWG)	Logic/sensor wiring	1	10	10
Wiring	Power Wire (16–18 AWG)	Battery/motor power wiri	1	12	12
Wiring	Connector Kit (JST/Dupont 560 pcs)	Modular wiring connector	1	24.95	24.95
Wiring	Heat-Shrink Tubing	Wire insulation and protec	1	7	7
Wiring	Ferrite Beads (Clip-On, 4pcs)	Reduce electrical noise	4	2	8
Interfaces	USB-UART Bridge (CP2102N / FT232)	Serial debugging & micro-	1	5.95	5.95
Interfaces	Wire & Safety Labels	Clear harness labeling	1	5	5
Net Total	**840.89**		**Total + Tax (8%)**		**908.1612**

MCU	119.7	$129.28	10.8%	1200
Interfaces	70.95	$76.63	6.4%	1200
Sensing	309.95	$334.75	27.9%	1200
Drivetrain	85	$91.80	7.7%	1200
Power	115.37	$124.60	10.4%	1200
Chassie	71.97	$77.73	6.5%	1200
Wiring	67.95	$73.39	6.1%	1200
Leftover	0	$291.84		



Component	Function / Problem Solved
Raspberry Pi 5 (8GB)	Acts as the main controller running ROS 2 for SLAM and navigation. Processes LiDAR, camera, and IMU data for map generation and decision-making.
Teensy 4.0	Handles motor PWM, encoder feedback, and IR sensor inputs. Provides precise timing and control, offloading low-level tasks from the Raspberry Pi.
RPLIDAR A2M12	Creates a 2D map of the environment for SLAM. Detects obstacles and provides accurate range data for localization and path planning.
Adafruit BNO055 IMU	Provides orientation and acceleration data for odometry input into SLAM. Works with the camera and sound sensors to correct drift in position and heading.
RPi Camera Module 3 Wide (IMX708)	Enhances map stability and SLAM accuracy, additional fault system for SLAM map
Ultrasonic Ranger Sensors	Fault system for SLAM processing so the map doesn't shift
FIT0186 Motors with Encoders (×2)	Provides differential drive motion and encoder feedback for odometry, enabling accurate position tracking during SLAM operations.
L298N Motor Driver (4 Pack)	Controls direction and speed of the drive motors via the Teensy. Converts control signals into motor output for smooth differential navigation.
TalentCell 12 V 26,400 mAh Battery	Powers the entire system, including the Pi, LiDAR, and motors. Provides stable voltage and long runtime for continuous SLAM operation.

RPLiDAR A2M12: Specification	**Value**
Type	2D Rotating Scanning LiDAR (Mechanical)
Cost	$229 USD
Range	0.2 – 12 meters
Accuracy	±2 cm
Update Rate	5 – 15 Hz (adjustable)
Sampling Rate	~16,000 samples/sec
Field of View	360° (Horizontal)
Interface	USB / UART
Power	5 V @ 500 mA
ROS / SLAM Support	Official rplidar_ros and rplidar_ros2 packages
Best Use Case	Full 2D SLAM Mapping for Indoor Mobile Robot
Notes	Excellent stability, mature SDKs, and low power draw.

Category	Part	Model & Link	Price	Quanity	Total
Chassie	Aluminum Extrusions	4pc 400mm	18.99	1	
Chassie	Extrusion Nuts and Bolts	200pc	9.99	1	

Category	Part / Component	Description	Quantity	Unit Price (USD)	Total (USD)
Sensing	Base Hat (103030275)	Sensor interface expansion	1	10.9	10.9
Power	Logic Level Shifter (Adafruit 5637)	Voltage level shifting for logic	3	2.95	8.85
Power	UBEC Buck Converter	12V to 5V regulator	1	9.95	9.95
Power	16 AWG Inline Fuse	Power line protection	1	8.99	8.99
Power	DC Barrel Connector	Power connection adapter	1	5.99	5.99
Power	XT60 PDB Power Hub	Power distribution board	1	9.6	9.6
Chassis	Aluminum Extrusions 400mm (4pc)	Structural frame for robot	1	18.99	18.99
Chassis	Extrusion Nuts and Bolts (200pc)	Fasteners for assembly	1	9.99	9.99
Drivetrain	Wheels & Hubs (90mm rubber)	Traction wheels for motor shafts	2	12	24
Drivetrain	Caster Wheel (50mm swivel)	Third-point balance support	1	8	8
Safety	Emergency Stop (Mushroom Switch)	Cuts motor power instantly	1	10.22	10.22
Power	Master Power Switch (High-Current)	Main system on/off switch	1	8	8
Power	12.6V Li-ion Smart Charger	Battery recharging unit	1	18	18
Mechanical	Dual-Slot Mail Bucket (3D Printed)	Holds and secures small parcels	1	20	20
Mechanical	Zip Ties & Cable Clips	Cable organization and strain relief	1	6	6
Wiring	Signal Wire (22–24 AWG)	Logic/sensor wiring	1	10	10
Wiring	Power Wire (16–18 AWG)	Battery/motor power wiring	1	12	12
Wiring	Connector Kit (JST/Dupont 560 pcs)	Modular wiring connectors	1	24.95	24.95
Wiring	Heat-Shrink Tubing	Wire insulation and protection	1	7	7
Noise Control	Ferrite Beads (Clip-On, 4pcs)	Reduce electrical noise	4	2	8
Interfaces	USB-UART Bridge (CP2102N / FT232)	Serial debugging & micro-ROS comms	1	5.95	5.95
Interfaces	Wire & Safety Labels	Clear harness labeling	1	5	5
	TOTAL ESTIMATED COST				846.12

Resource Costs				
Title	Yearly Salary	Hourly Rate	Billable Hours	Amount Owed
Project manager	$80,000.00	$38.50	95	$3,657.50
Mechanical Engineer	$75,000.00	$36.00	97	$3,492.00
Software Engineeer 1	$82,000.00	$39.50	94	$3,713.00
Software Engineeer 2	$82,000.00	$39.50	94	$3,713.00
Hardware Engineer	$78,000.00	$37.50	95	$3,562.50
			Total Amount	$18,138.00